

Faculty of Science & Technology
PES's Modern College of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Level 7: First Year M. Tech. (2024 Pattern)

Course Code	CSE08651
Course Name	Skill Development – II
Semester	IV
Experiment No.	7
Title	Input Data Analysis with Python using Jupyter Notebook

Problem Statement

To study and analyze input research data using Python programming tools for performing data preprocessing, exploration, and analysis.

Objective

To make use of Python and Jupyter Notebook for performing exploratory data analysis and understanding the characteristics of input datasets.

Outcome

CO653.7: Apply Python-based tools to analyze and interpret research data effectively.

Software / Hardware Requirements

- Python Programming Environment
- Jupyter Notebook
- Data Analysis Libraries (NumPy, Pandas, Matplotlib)
- Personal Computer or Laptop

Theory

Input data analysis is a crucial step in the research and data science workflow, as it involves examining, cleaning, and understanding raw data before applying advanced analytical or machine learning techniques. Proper data analysis helps identify patterns, trends, anomalies, and data quality issues, which directly influence the reliability of research outcomes.

Python is a widely used programming language in data analysis due to its simplicity, flexibility, and extensive ecosystem of libraries. Libraries such as NumPy and Pandas provide efficient data structures and functions for numerical computation and data manipulation, while visualization libraries enable graphical representation of data for better interpretation.

Jupyter Notebook is an interactive computing environment that allows users to combine code, visualizations, and explanatory text in a single document. It supports iterative data exploration and facilitates reproducible research. By using Python with Jupyter Notebook, researchers can perform systematic input data analysis and document insights effectively.

Steps Involved

1. Jupyter Notebook was launched in the Python environment.
2. The MNIST dataset was loaded using Python libraries.
3. Data preprocessing operations such as normalization were performed.
4. Exploratory data analysis was conducted using statistical summaries.
5. Visualizations were generated to understand pixel distributions.

Input / Dataset Details

Name of the Dataset: MNIST Handwritten Digits Dataset

Description of the Dataset: The MNIST dataset consists of grayscale images of handwritten digits ranging from 0 to 9. It is widely used for evaluating image processing and machine learning algorithms, particularly in digit recognition tasks.

Dataset Characteristics:

- Image-based dataset

- Labeled supervised learning dataset
- Balanced class distribution

Subject Area: Computer Vision, Machine Learning, Pattern Recognition

Associated Tasks: Digit recognition, image classification, feature analysis

Feature Type: Numerical (pixel intensity values)

Number of Instances: 70,000 images (60,000 training and 10,000 testing samples)

Number of Features: 784 features per instance (28×28 pixel images)

Results

The MNIST dataset was successfully analyzed using Python in Jupyter Notebook. Key characteristics of handwritten digit images were identified through statistical analysis and visualization techniques.

Analysis and Conclusion

Analysis: Exploratory analysis of the MNIST dataset revealed consistent pixel intensity patterns across digit classes. Visualization techniques helped in understanding digit structure and data distribution.

Conclusion: Input data analysis using Python and Jupyter Notebook provided valuable insights into the MNIST dataset. Such analysis is essential for preparing image data for digit recognition and machine learning applications.

References

1. LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P., “Gradient-Based Learning Applied to Document Recognition,” Proceedings of the IEEE.
2. Jupyter Project, “Jupyter Notebook Documentation,” Available: <https://jupyter.org>